# BMW Sales & Market Analysis

Global Vehicle Sales · 2010–2024

01

# Business Understanding

# Project & Client Introduction

Our client, BMW, is a global leader in premium mobility, focused on "intelligence and premium mobility." The core business involves the manufacturing and sales of high-performance luxury vehicles.

| | |
|---|---|
| **Client** | Bayerische Motoren Werke Aktiengesellschaft (BMW) |
| **Business** | Global leader in the automotive industry (high-performance & luxury vehicles). |
| **Client Need** | Improve overall business/sales by understanding market demand, regional strengths/weaknesses, and features that drive sales. |

# Case study:
# BMW Sales & Market Analysis (2010-2024)

**Primary Purpose:** Analyze BMW's historical sales and vehicle data (2010–2024) to uncover meaningful market insights and descriptive patterns in product performance and customer demand.

# Introduction

- This project examines Bayerische Motoren Werke Aktiengesellschaft’s (BMW) historical sales and vehicle data to identify trends in product performance and customer demand across regions and models.
- Using data analysis and predictive techniques, it aims to uncover meaningful market insights, forecast future sales, and segment markets based on key vehicle and sales characteristics.
- Findings will be communicated through visualizations and recommendations.
- The scope is limited to BMW’s internal historical data and does not incorporate real-time data or external economic factors.

# Business Questions

1. Can we predict 2025 car sales volume using past yearly totals?

2. How can BMW better understand its customers by grouping vehicles based on technology preference rather than price alone?

3.Can Vehicle Specifications predict high sales volume?

# 5W 1H Analysis
# Who

| # | Question | Answer |
|---|---|---|
| Q1.1 | Who is involved? | BMW company, consumers, the global automotive market, automotive industry |
| Q1.2 | Who is affected? | BMW company, consumers, the global automotive market, automotive industry |
| Q1.3 | Who will benefit? | BMW company, consumers, the global automotive market, automotive industry |
| Q1.4 | Who will be harmed? | Competitors - Lexus, Audi, and Acura |

# 5W 1H Analysis
# What

| # | Question | Answer |
|---|---|---|
| Q2.1 | What is your topic narrowed down in a simple phrase/sentence? | Using data modelling to gain insight on where BMW's weaknesses and strengths lie in the automotive market to improve sales. |
| Q2.2 | What does your topic involve? (i.e. What are the different parts to it?) | This topic involves analyzing BMW's vehicle data to understand and predict strategic business insights. |
| Q2.3 | What is it similar to / different from? | This topic is similar to other luxury car manufacturers in the world. |
| Q2.4 | What might be affected/changed by your topic? | The market strategies , customer targeting and sales planning of BMW might be affected by this topic |

# 5W 1H Analysis
# When

| # | Question | Answer |
|---|---|---|
| Q3.1 | When does this take place? When did this take place? When will it take place? When should this take place? | The data set is from the year 2010-2024. It will continue through the forecasting stage , projecting BMW's sales and pricing trends. |
| Q3.2 | Does when this takes place affect the topic? | Yes, the past data explains the trends and the future forecasts guides BMW's business strategy. |

# 5W 1H Analysis
# Where

| # | Question | Answer |
|---|---|---|
| Q4.1 | Where does this take place? (Where did it …. Where will it … Where should it ….?) | The analysis takes place across North America, South America, Asia, Africa and the middle east.<br>It reflects past BMW sales activities in these regions.<br>Forecasting will project future trends in these regions.<br>This type of analysis should be conducted regularly across all markets |
| Q4.2 | Does it matter where it takes place? Is it affected by location? | Location affects the topic because BMW's sales, pricing and customer preferences vary significantly across regions such as North America, Europe, Asia, Africa, South America and the Middle East. |

# 5W 1H Analysis
# Why

| # | Question | Answer |
|---|---|---|
| Q5.1 | Why is this topic important? Why does it matter? | The topic is important for BMW to understand their consumers, sales and trend patterns, the automotive market/industry. It can be beneficial for the BMW company to make business decisions. |
| Q5.2 | Why do certain things happen? (What are some causes and effects within the topic?) | Sales is dependent on region, mileage, model, color, etc |

# 5W 1H Analysis
# How

| # | Question | Answer |
|---|---|---|
| Q6.1 | How does this topic work? How does it function? How does it do what it does? | BMW's global sales system works by matching supply with demand. Data from past sales helps BMW forecast future demand and plan production. |
| Q6.2 | How did it come to be? | High sales volume emerge because of consumer demand shaped by competitive pricing, performance and strong brand reputation. |
| Q6.3 | How are those involved affected? | BMW data affects the company, employees, customers and the investors |

# 5W 1H Analysis Summary

| Question | Answer |
|---|---|
| **Who** is involved? | Bayerische Motoren Werke AG (BMW), consumers, the global automotive market, automotive industry, competitors. |
| **What** is your topic narrowed down in a simple phrase/sentence? | Using data modelling to gain insight on where BMW's weaknesses and strengths lie in the automotive market to improve sales. |
| **When** does this take place? | The dataset is from the years 2010-2024. |
| **Where** does this take place? | The analysis focuses on BMW sales data from Africa, Asia, Europe, the Middle East, North and South America. |
| **Why** is this topic important? **Why** does it matter? | The topic is important for BMW to understand their consumers, sales and trend patterns, the automotive market/industry. It can be beneficial for the BMW company to make business decisions. |
| **How** does this topic work? **How** does it function? **How** does it do what it does? | BMW's global sales system works by matching supply with demand. Data from the past sales helps BMW forecast future demand and plan production. |

# STRATEGIC VALUE PROPOSITION

### INFORMED DECISIONS

Providing actionable insights that support high-level decision making and long-term planning for the company.

### EFFICIENCY

Enabling engineers to design future models based on data-proven popular features that connect to higher sales.

### REVENUE GROWTH

Allowing the Marketing Team to optimize campaigns by deeply understanding consumer preferences.

### FORECASTING

Predicting future trends to enable strategic planning for sustainable growth and inventory management.

# CRITICAL PROCESSES & ORGANIZATIONAL IMPACT

| CRITICAL PROCESS | IMPACT & BENEFIT | STAKEHOLDER |
|---|---|---|
| MARKET ANALYSIS | Segment cars into price-based categories to identify specific customer preferences across segments. | Marketing Team |
| FINANCIAL FORECASTING | Develop predictive models for future sales (e.g., 2025 volume) to support resource allocation. | BMW Company |
| PRODUCT INNOVATION | Highlight specific features and models that drive sales to inform future engineering designs. | BMW Engineers |
| CRM OPTIMIZATION | Leverage consumer preference data to optimize communication and marketing efforts. | Marketing Team |

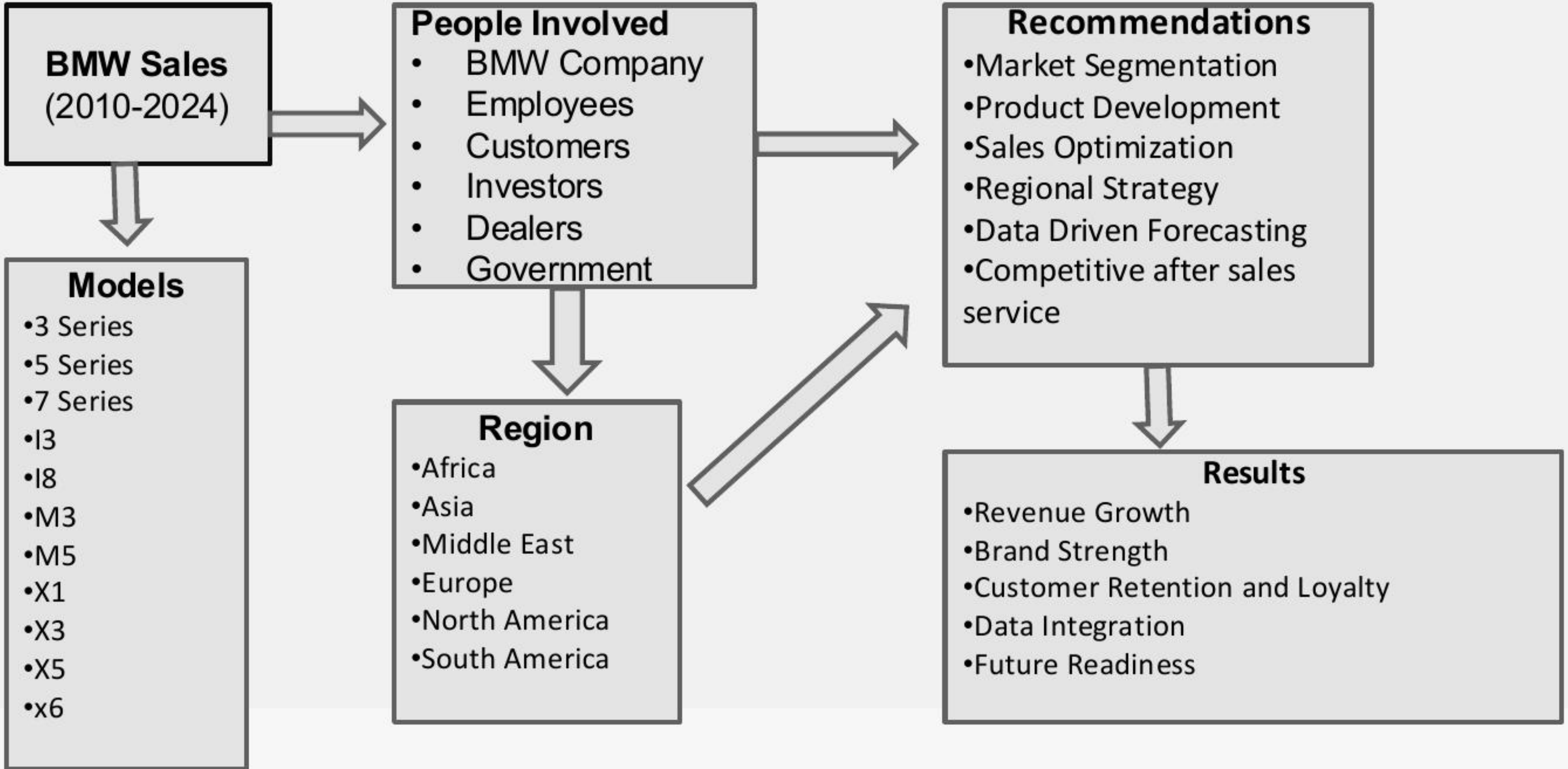
02

# Data Understanding

# Data Dictionary

| Field Name | Description | Data Type |
| --- | --- | --- |
| **Model** | The specific model of the BMW vehicle sold | Text / Categorical |
| **Year** | The year the vehicle was sold. | Integer |
| **Region** | The geographical region of the sale | Text / Categorical |
| **Color** | Th color of the vehicle. | Text / Categorical |
| **Fuel_Type** | The type of fuel source for the vehicle | Text / Categorical |
| **Transmission** | The vehicle's transmission type ('Manual' or 'Automatic'). | Text / Categorical |
| **Engine_Size_L** | Size of the vehicle's engine (liters) | Float |
| **Mileage_KM** | Total mileage recorded on the odometer in kilometers at the time of sale. | Integer |
| **Price_USD** | Final sale price of the vehicle in US Dollars. | Integer |
| **Sales_Volume** | Total number of units of this specific Model/Year/Region combination sold. | Integer |
| **Sales_Classification** | Indicating the sales success of the record ('High' or 'Low'). | Text / Categorical |

# Data Modeling – Conceptual Model

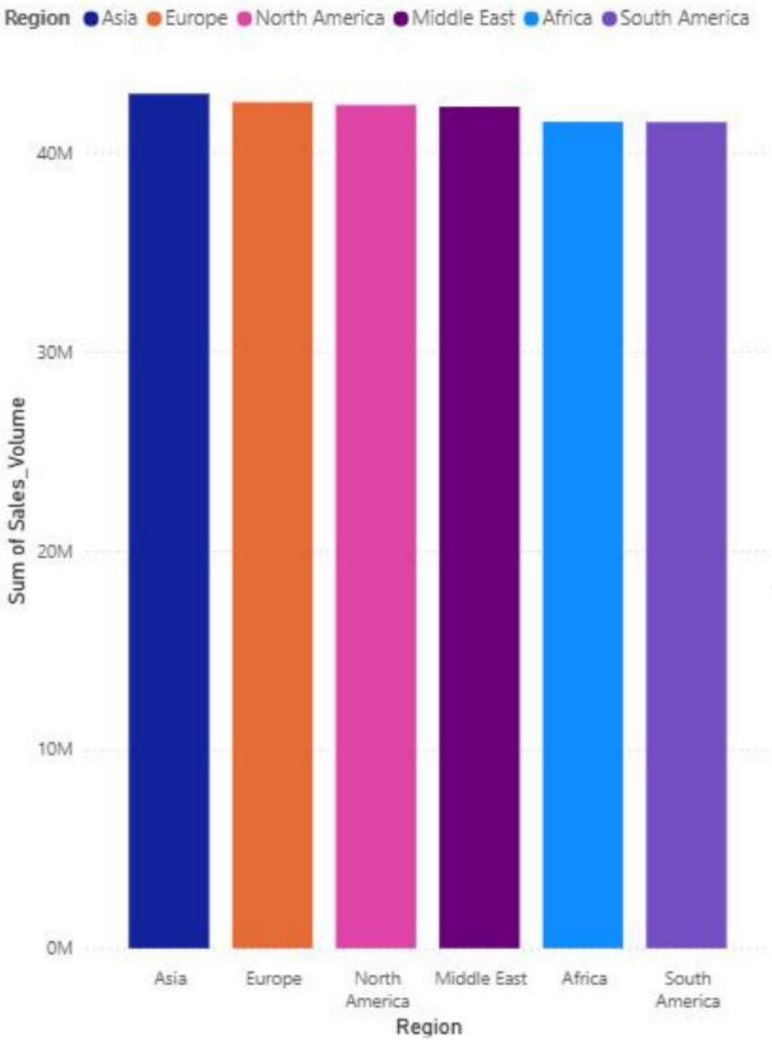


# Data Preparation

| Model | Year | Region | Color | Fuel_Type | Transmission | Engine_Size_L | Mileage_KM | Price_USD | Sales_Volume | Sales_Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 Series | 2016 | Asia | Red | Petrol | Manual | 3.5 | 151748 | 98740 | 8300 | High |
| i8 | 2013 | North America | Red | Hybrid | Automatic | 1.6 | 121671 | 79219 | 3428 | Low |
| 5 Series | 2022 | North America | Blue | Petrol | Automatic | 4.5 | 10991 | 113265 | 6994 | Low |
| X3 | 2024 | Middle East | Blue | Petrol | Automatic | 1.7 | 27255 | 60971 | 4047 | Low |
| 7 Series | 2020 | South America | Black | Diesel | Manual | 2.1 | 122131 | 49898 | 3080 | Low |
| 5 Series | 2017 | Middle East | Silver | Diesel | Manual | 1.9 | 171362 | 42926 | 1232 | Low |
| i8 | 2022 | Europe | White | Diesel | Manual | 1.8 | 196741 | 55064 | 7949 | High |
| M5 | 2014 | Asia | Black | Diesel | Automatic | 1.6 | 121156 | 102778 | 632 | Low |
| X3 | 2016 | South America | White | Diesel | Automatic | 1.7 | 48073 | 116482 | 8944 | High |

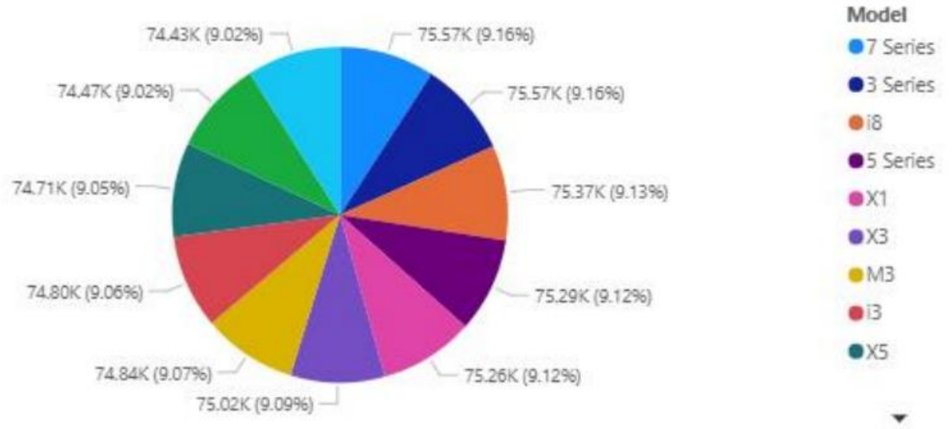
The dataset was retrieved from Kaggle. Since it was already clean, it was uploaded directly to the workspace.

## Sum of Sales_Volume by Region and Region




## Regions Represented in BMW 2010-2024 Sales Data


## Average of Price_USD by Model



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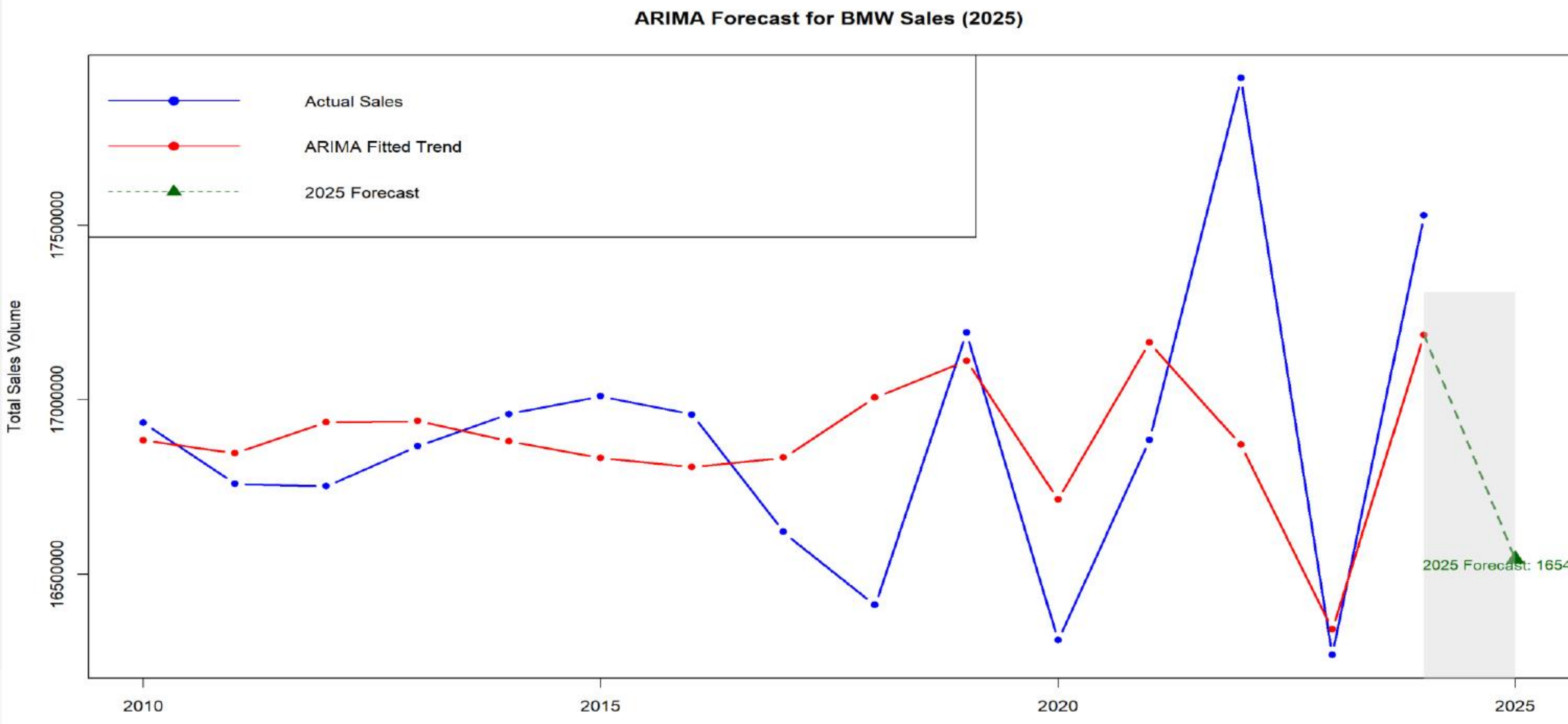
# Modelling

# Predicting BMW Sales Volume (Forecasting)

Can we predict 2025 car sales volume using past yearly totals?

# Visual 1 – Forecasting



# ➢ Why ARIMA

- The data is highly variable, not a simple trend
- ARIMA captures time-based patterns
- Provides both predictions and uncertainty
- Auto.ARIMA selects the best model automatically

# Interpretation 1

- ➢ **Blue Line** → Actual BMW sales (2010–2024)
- Indicates the series is not purely trending, making ARIMA a suitable model.

- ➢ **Red Line** → ARIMA fitted values
- Represents how the ARIMA model reconstructs past sales.

- **Green Triangle** → ARIMA 2025 Forecast
- The model predicts BMW sales for 2025 using detected historical patterns.
- This point is the best estimate generated by the ARIMA model.

- ➢ **Prediction Interval** (Gray Shaded Area)

- The shaded band represents the uncertainty around the 2025 forecast.
- The upper and lower limits come from ARIMA’s 95% confidence interval.

# ……Continued

➢ The wide interval indicates:

- The model detects high volatility in past sales.
- Forecasting beyond one year carries greater uncertainty.
- ARIMA provides a range, not just a single number, to reflect this uncertainty.

➢ **2025 ARIMA Forecast (= 16543085)**

- The ARIMA model predicts 16.54 million for BMW sales in 2025.
- This value represents the expected sales based on historical patterns from 2010–2024.
- The forecast is shown as a single estimated point, with uncertainty captured in the prediction interval.

# Recommendations

- **Prepare for multiple demand scenarios**

Because the prediction interval is wide, BMW should plan for different situations.

- **Strengthen forecasting with more detailed data**

Since the model is based on yearly totals, its accuracy would increase by incorporating other influencing variables.

- **Improve inventory and Supply chain flexibility**

BMW should stay flexible by keeping supply chains agile and adjusting production to real-time demand.

- **Monitor market trends closely in 2025**

BMW should pay attention to shifting consumer demand, regional sales performance, economic conditions, and global supply-chain disruptions.
This allows BMW to react quickly to unexpected changes.

# Recommendations

- **Use predictive analytics for decision-making**

The analysis shows the value of forecasting, helping BMW integrate key data and apply predictions to budgeting, marketing, and production planning.

**Summary**

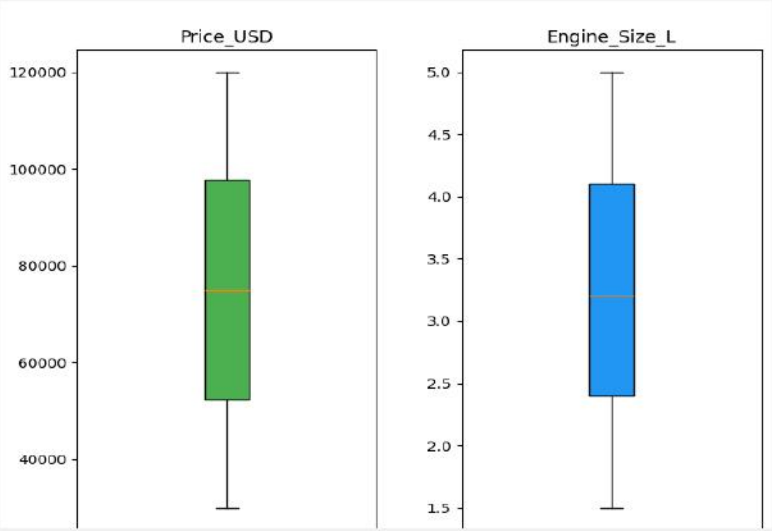
- BMW should prepare flexible strategies for 2025, monitor market trends closely, and enhance forecasting by integrating more detailed and external data due to the high volatility seen in past sales.

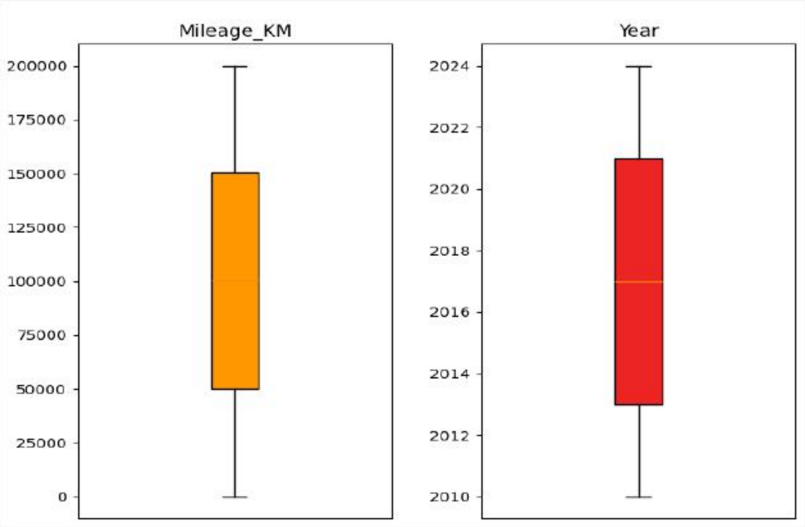
# Conclusion

- The ARIMA model successfully forecasted BMW's total sales volume for 2025 using historical yearly data from 2010–2024. The forecast predicts a total sales volume of 16,543,085, indicating a potential increase in sales activity for the upcoming year. However, the wide prediction interval highlights that BMW's past sales patterns have been highly variable, meaning the exact 2025 outcome remains uncertain.

- Overall, the model suggests growth is likely, but volatility remains a key factor, requiring BMW to plan for multiple sales scenarios. This forecast provides valuable insight for strategic planning, emphasizing the importance of flexible production, market monitoring, and enhanced forecasting models using more detailed data in the future.

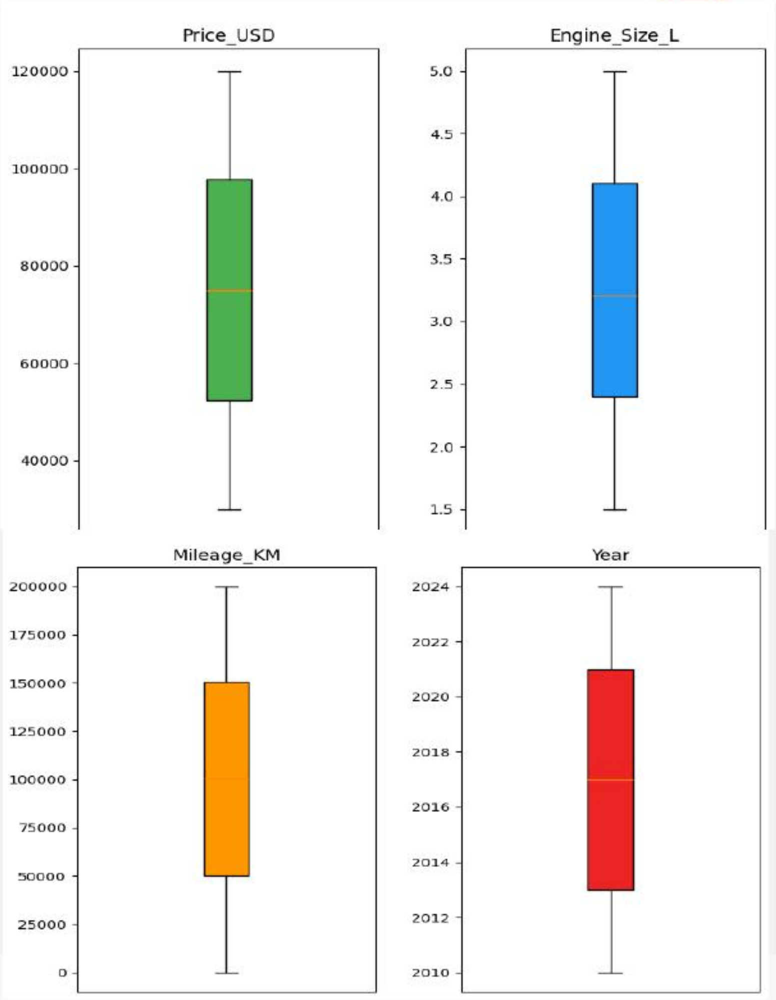
# Visual 2 – K Means



## Why Multi-Feature Segmentation Was Required

- Price-only clustering created groups with almost identical averages in price, mileage, and model year.
- BMW vehicles in this dataset fall within a narrow premium price band, making price ineffective for separating customers.
- Multi-feature segmentation included fuel type, transmission, region, and model in addition to numeric variables.
- Encoding and standardizing these features allowed K-Means to segment customers by technology preference and usage style, not just how much they paid.



# Results of K-Means Clustering (K = 3)

- Clustering revealed three distinct customer segments, driven primarily by fuel and technology preference rather than price or engine size.
- Cluster 0 – Diesel/Hybrid Transition Segment: A near 50/50 mix of diesel and hybrid vehicles, representing customers focused on fuel efficiency and gradual technology transition.
- Cluster 1 – Full Electric Segment: Comprised entirely of EVs, indicating customers ready for full adoption of modern electric technology across global regions.
- Cluster 2 – Traditional Petrol Segment: 100% petrol vehicles, reflecting customers who prefer conventional driving performance and familiar ownership models.

Numeric summary by Cluster:

| Cluster | Price_USD | Engine_Size_L | Mileage_KM | Year |
|---|---|---|---|---|
| 0 | 74936.1 | 3.2 | 100477.0 | 2017.0 |
| 1 | 75276.3 | 3.2 | 100524.3 | 2017.0 |
| 2 | 74990.4 | 3.3 | 99753.5 | 2017.0 |

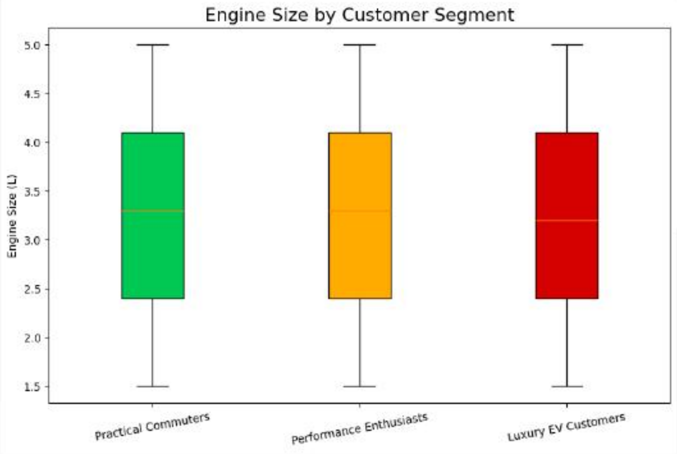
Fuel-type distribution by Cluster (row-normalised):

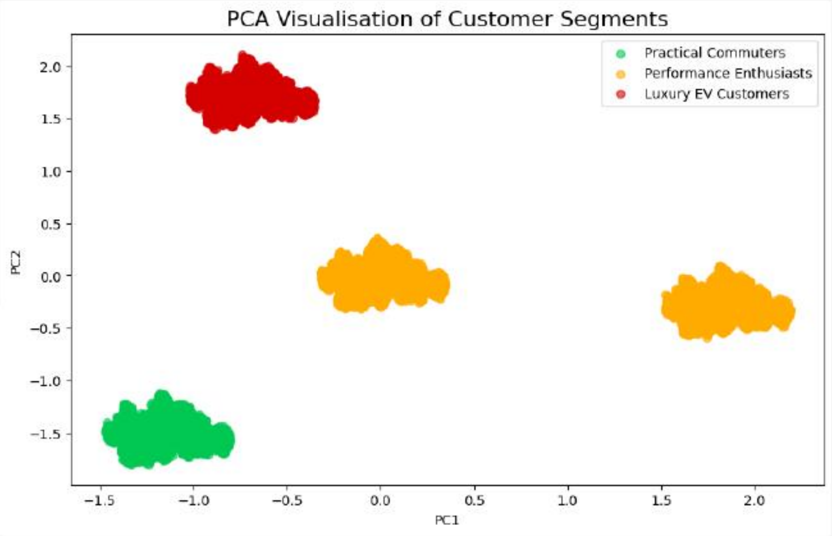
| Fuel_Type / Cluster | Diesel | Electric | Hybrid | Petrol |
|---|---|---|---|---|
| 0 | 0.49 | 0.0 | 0.51 | 0.0 |
| 1 | 0.00 | 1.0 | 0.00 | 0.0 |
| 2 | 0.00 | 0.0 | 0.00 | 1.0 |

Region distribution by Cluster (row-normalised):

| Region / Cluster | Africa | Asia | Europe | Middle East | North America | South America |
|---|---|---|---|---|---|---|
| 0 | 0.16 | 0.17 | 0.17 | 0.17 | 0.17 | 0.16 |
| 1 | 0.17 | 0.16 | 0.17 | 0.16 | 0.17 | 0.17 |
| 2 | 0.17 | 0.17 | 0.16 | 0.17 | 0.16 | 0.17 |

# Price and Engine Size Clustering

# What does this mean for BMW

Findings confirm that price and performance do not differentiate BMW customers within this dataset.

Fuel-type segmentation shows clear separation:

- Cluster 2: 100% petrol drivers (traditional buyers).
- Cluster 1: 100% electric (full technology adopters).
- Cluster 0: Diesel–hybrid mix (transition-stage customers).

- PCA visualization validates these distinctions, showing three clearly formed behavioral groups.

- Business strategy should focus on technology adoption readiness, not price sensitivity.

Marketing messages should align with segment needs:

- Traditional performance, efficiency and cost savings, or sustainability and innovation.

- Using multi-feature segmentation was essential — it reveals how BMW customers differ in their journey toward newer vehicle technology, not just in how much they spend.

# Recommendations

**Target marketing based on technology readiness**

- Petrol drivers: emphasize traditional performance and reliability.
- Hybrid/diesel drivers: highlight fuel savings and gradual technology transition.
- Electric drivers: focus on sustainability, charging convenience, and connected features.

**Adjust product planning and inventory to match demand**

- Maintain petrol inventory where newer technology adoption is slower.
- Expand hybrid options for transitioning customers.
- Grow EV models and support infrastructure in regions showing widespread adoption.

**Support customers as they move toward newer technology**

- Offer cost-comparison tools and upgrade pathways (petrol → hybrid → electric).
- Provide education on charging, maintenance, and long-term performance.
- Introduce incentives that reduce adoption barriers and encourage confidence.

# Conclusion

- Multi-feature K-Means showed that **technology preference, not price** best explains BMW customer differences in this dataset.

- Although all segments fall within a similar premium price range, **fuel-type choices** (petrol, hybrid/diesel, electric) reveal distinct behavioral patterns.

- The analysis confirms that **price-only segmentation** would group fundamentally different customers together, missing key insights into their readiness for newer vehicle technology.

# Predicting BMW Sales Volume(Classification)

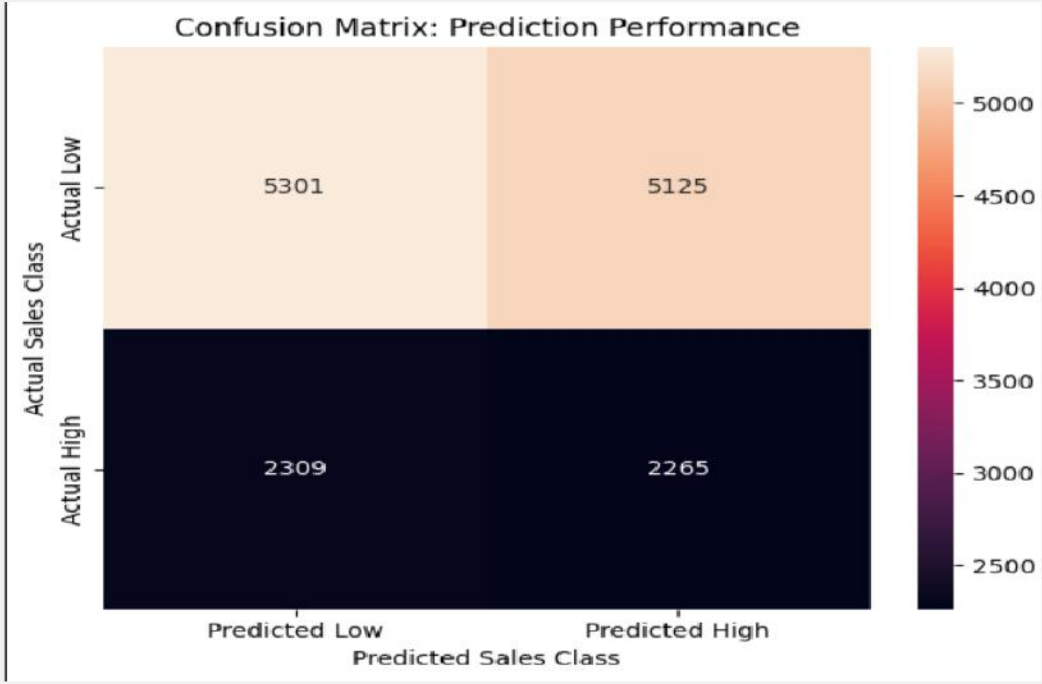
Can Vehicle Specifications predict high sales volume?

# Methodology: Ensuring Model Integrity

Main points :

- **Goal:** Predict if a car is "High Sales" or "Low Sales" (Binary Classification).
- **Model:** Logistic Regression with class_weight (parameter) = ' Balanced '
- **Removed:** `Sales_Volume` column.
- **Reason:** Prevents "Data Leakage"
- **Result:** The model must rely **only** on vehicle attributes (Model, Year, Color).

# Visual 3 - Classification



# Principal findings

- **Accuracy: 50.4%**
- **Visual Analysis:**
- The Visual Evidence of Indiscriminate Classification pattern shows the model is confused.
- False Positives (Predicted High, Actual Low) are very high.
- **Finding:** The model cannot distinguish between High and Low sales.

# Business Conclusion

**Conclusion:**

- Vehicle specifications (Engine, Color, Model) have **zero statistical impact** on sales volume.
- *Evidence:* Model accuracy is 50.4%, identical to random chance.

**Strategic Recommendation:** Analytics focus to Market data :

- **Competitor Pricing:** Analyze price gaps with Audi/Mercedes.
- **Marketing Spend:** Correlate regional advertising spend with volume.
- **Macroeconomics:** Track interest rates and GDP growth.

# Project Conclusion

The main takeaway is: To sell more cars, BMW should focus on who the customer is, not just what the car has.

The Visuals show that:

- The market is volatile and hard to predict. (The sales forecast shows a wide range for future sales).
- Customers group together based on the type of fuel they want (like gas vs. electric), which is more important than the car's price.
- The specific features of a car (like horsepower or size) don't actually help predict if it will be a best-seller. (The prediction chart was almost 50/50, meaning the car features didn't matter for sales success).